ARTIFICIAL INTELLIGENCE IN ROBOTICS

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- Deterministic vs Intelligent Robotics
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Why AI in Robotics?

- **Real world problems:**
  - *Partial observability*
  - *Ambiguous perception*
  - *Dynamic environments*
  - *Unpredictable humans*
  - *Unknown objects*
  - *Long-horizon tasks*
- **Classical robotics struggles when:**
  - *The environment changes*
  - *The rules are incomplete*
  - *Models are inaccurate*

AI provides adaptation and learning.

Deterministic vs Intelligent Robotics

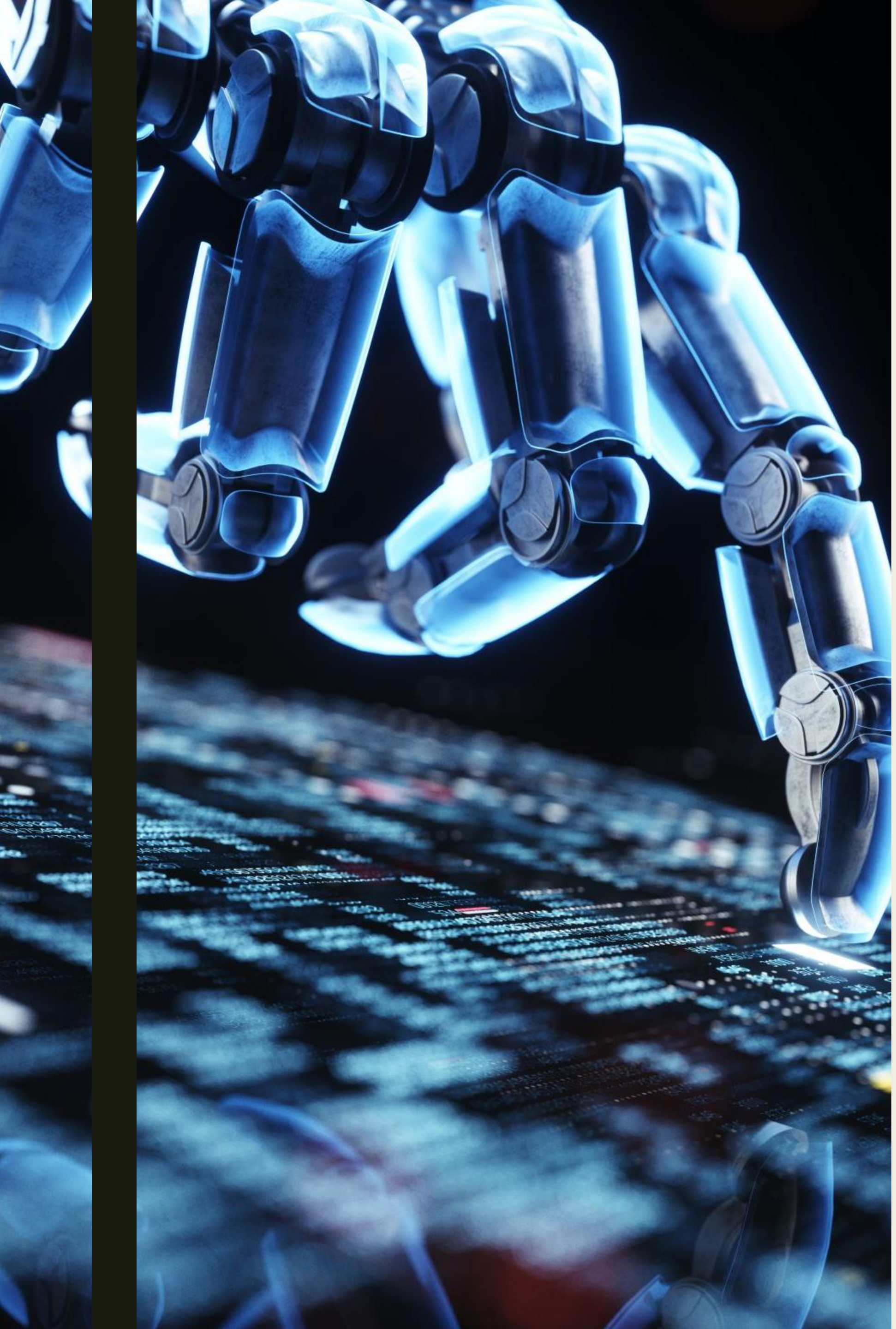
Deterministic	Intelligent
Classical Pipeline	AI-Augmented Pipeline
Predefined Rules	Learned Policies
Fixed Thresholds	Learned Features
Manually Tuned	Data-Driven
Model-based	Hybrid Model + Data
Limited Generalization	Scalable Generalization

AI allows robots to:

- Improve with experience
- Adapt to new environments
- Handle uncertainty probabilistically

Probabilistic Reasoning in Robotics

- Robots operate under uncertainty:
  - *Sensor noise*
  - *Model errors*
  - *Actuator drift*
  - *Human unpredictability*
- AI Frameworks for Uncertainty:
  - *Bayesian Inference*
  - *Markov Decision Processes (MDPs)*
  - *Partially Observable MDPs (POMDPs)*
  - *Belief state modeling*
- Key Concept:
  - *Robot maintains a probability distribution over possible states.*



Machine Learning in Robotics

- AI enables robots to learn:
  - *Perception models*
  - *Control policies*
  - *Behavioral strategies*
  - *Representations of the world*
- Machine Learning paradigms:
  - *Supervised learning*
  - *Reinforcement learning*
  - *Imitation learning*
  - *Self-supervised learning*

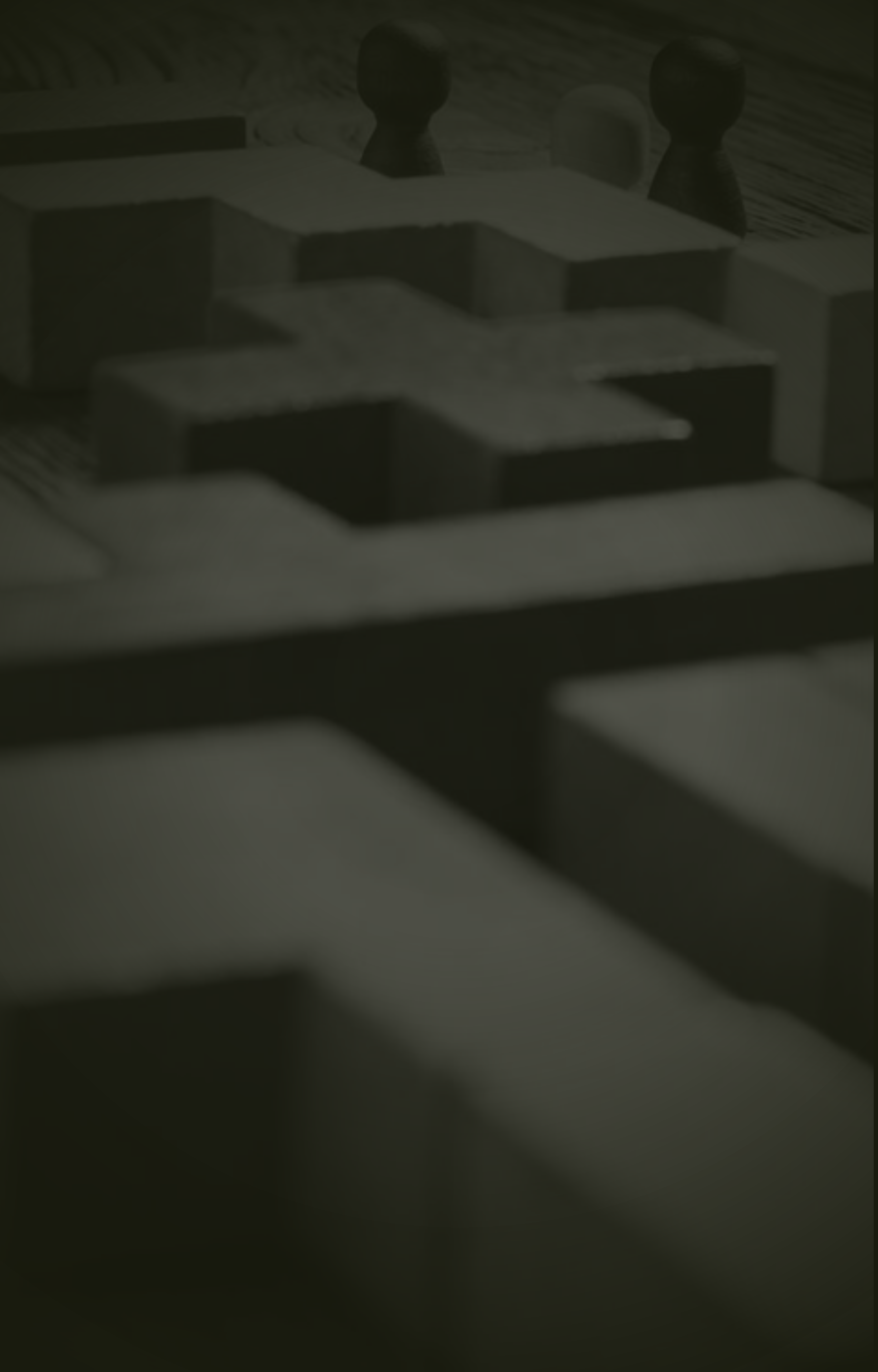
Reinforcement Learning in Robotics

RL Framework:

- Agent ↔ Environment
- State → Action → Reward → Next State

Why RL in Robotics?

- Hard to model dynamics
- Complex contact physics
- Long-horizon behavior
- Learning directly from interaction



Challenges of RL in Robotics

- Sample inefficiency
- Safety during exploration
- Sim-to-real gap
- High-dimensional state spaces
- Sparse rewards
- Solutions:
  - *Simulation pre-training*
  - *Domain randomization*
  - *Offline RL*
  - *Model-based RL*
  - *Curriculum learning*

Imitation Learning

Robot learns from demonstrations.

Approaches:

- *Behavior cloning*
- *Inverse reinforcement learning*
- *Dataset aggregation (DAgger)*

Applications:

- *Robotic surgery*
- *Manipulation*
- *Autonomous driving imitation*

Advantage: More stable and safer than pure RL.

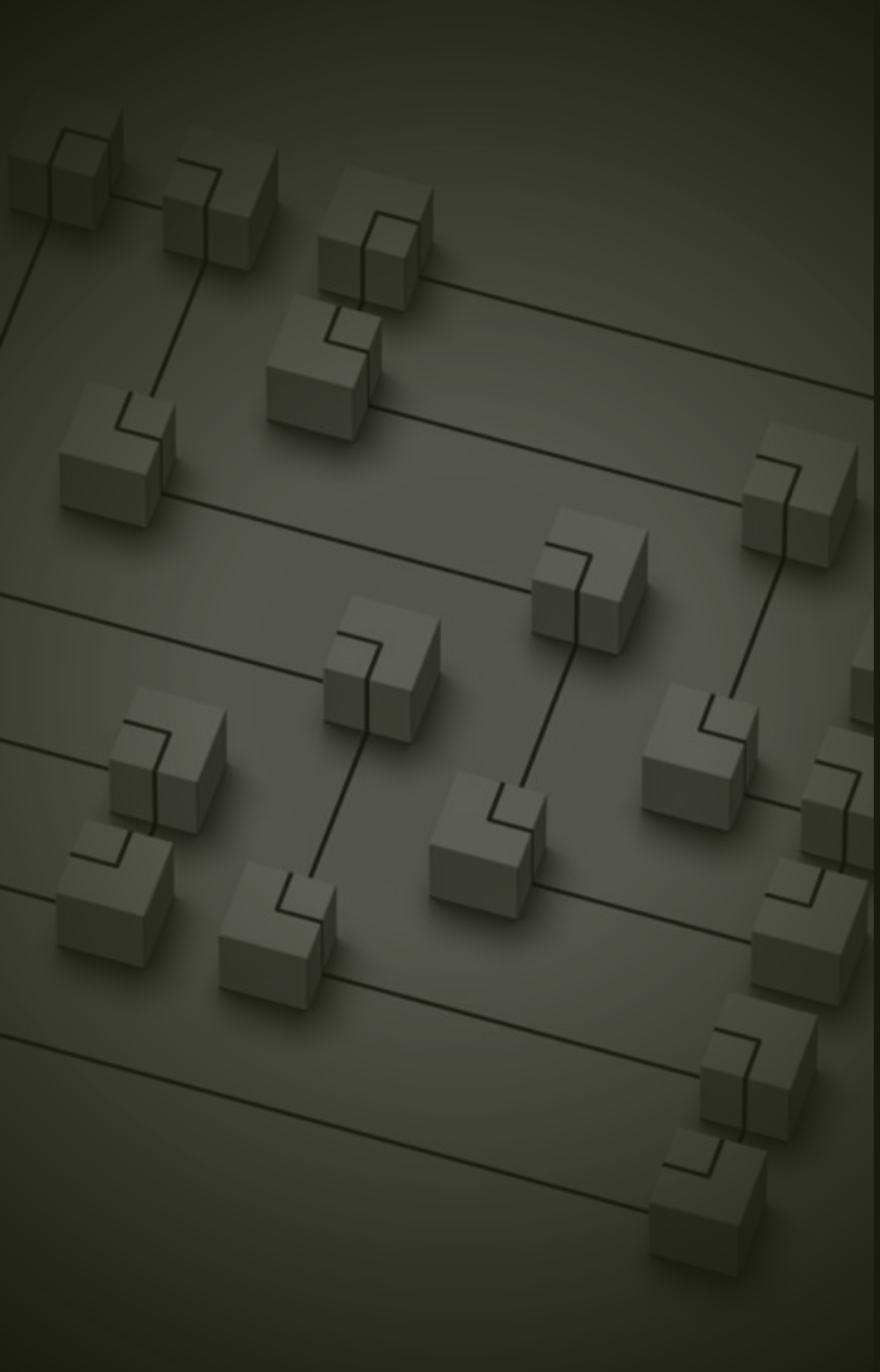
Self-Supervised Learning in Robotics

Robots generate their own labels.

Examples:

- *Predict next sensory input*
- *Learn visual embeddings*
- *Contrastive learning on video sequences*
- *Predict robot joint states*

Benefit: No manual labeling required.



Representation Learning

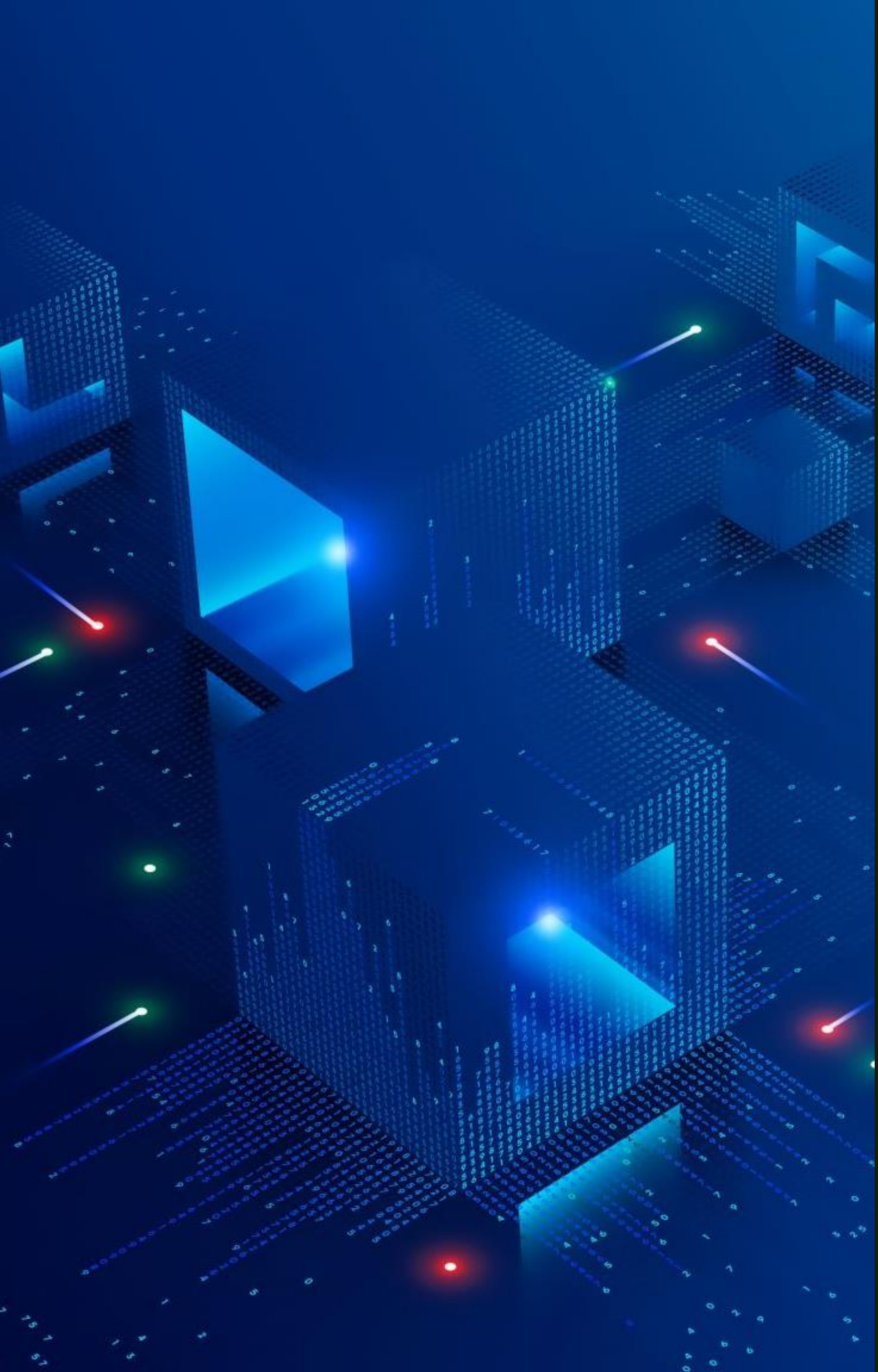
- Modern AI Robotics depends on good representations.
- Instead of raw pixels, Learn embeddings:
  - *Visual embeddings*
  - *Spatial embeddings*
  - *Object-centric representations*
  - *Latent state representations*
- Why important?
  - *Planning becomes easier*
  - *Generalization improves*
  - *Robustness increases*

Multi-Modal AI in Robotics

- Robots integrate:
  - *Vision*
  - *Language*
  - *Depth*
  - *Proprioception*
  - *Tactile sensing*
- Modern models:
  - *Transformers*
  - *Cross-attention architectures*
  - *Sensor fusion networks*

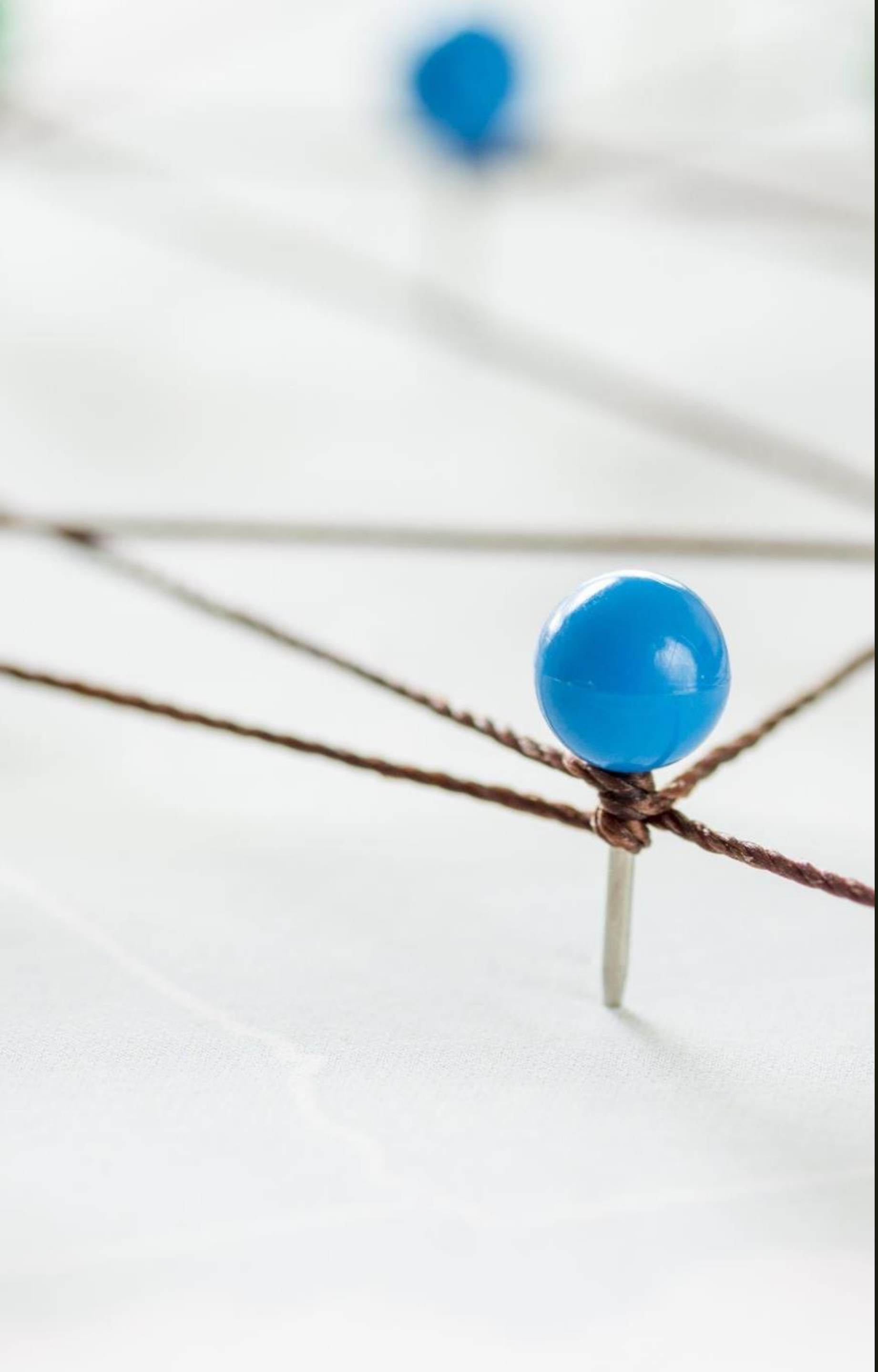
Foundation Models for Robotics

- Inspired by Large Language Models.
- What are they?
  - *Large-scale models trained on:*
  - *Massive image datasets*
  - *Robot demonstrations*
  - *Multi-task interactions*
- Capabilities:
  - *Zero-shot learning*
  - *Task generalization*
  - *Natural language control*



AI System Architecture in Modern Robotics

- High-level AI stack:
  - *Perception Embeddings → World Model → Policy Network → Planner → Controller*
- AI modules operate at:
  - *Semantic level*
  - *Task level*
  - *Decision level*



World Models

- AI learns a model of the environment.
- Instead of fixed maps, learn:
  - *Dynamics*
  - *Object interactions*
  - *Human behavior*
- Used for:
  - *Predictive planning*
  - *Long-horizon reasoning*
  - *Simulation-based imagination*

AI in Human-Robot Interaction

AI enables:

- Natural language understanding
- Gesture recognition
- Intention prediction
- Social behavior modeling

Robots become:

- Assistants
- Collaborators
- Companions

Safety & Reliability in Intelligent Robotics

AI introduces new risks:

- Overfitting
- Distribution shift
- Uncertainty miscalibration
- Adversarial inputs

Solutions:

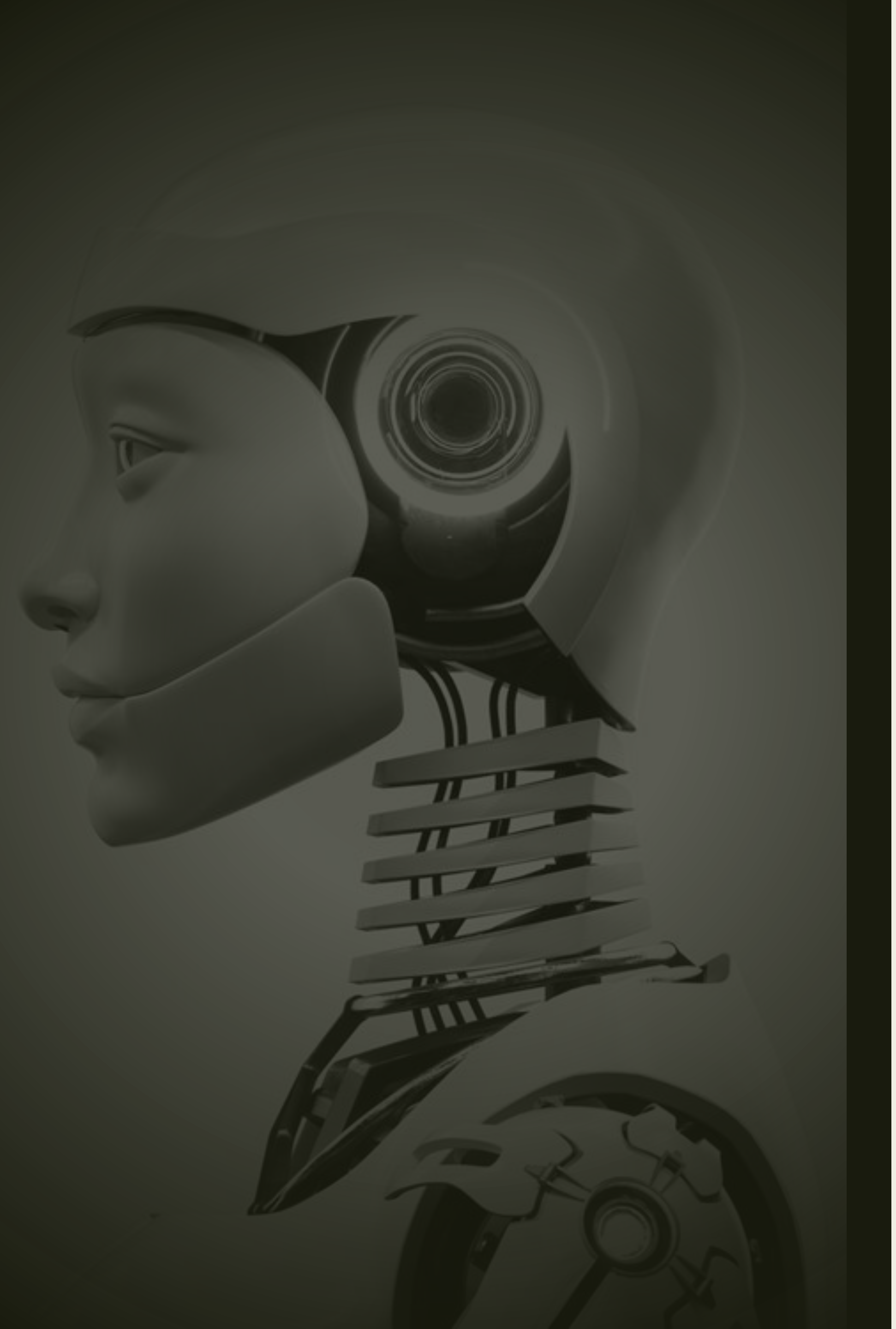
- Uncertainty estimation
- Ensemble models
- Formal verification
- Runtime monitoring
- Redundancy

Deployment Challenges

- Real-time constraints
- Edge computing limitations
- Hardware constraints
- Energy efficiency
- Data logging & retraining
- AI robotics must balance Accuracy vs Latency vs Safety

The Future of AI in Robotics

- End-to-end learning
- Embodied AI
- Lifelong learning robots
- Multi-agent systems
- General-purpose robotic intelligence
- Robotics foundation models



Summary

- AI enables adaptation beyond classical robotics.
- Probabilistic reasoning handles uncertainty.
- Reinforcement and imitation learning enable complex behaviors.
- Representation learning is foundational.
- Foundation models are shaping next-generation robotics.
- Safety and generalization remain open challenges.